Flux blindness in neural solvers for heterogeneous subsurface flow: Spectral mechanisms, pre-training screening, and flux-informed learning

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Abstract

Deep-learning surrogates for subsurface flow are conventionally trained and benchmarked on hydraulic head h, whereas contaminant transport and geological storage depend on the Darcy velocity $\mathbf{q} = -K\nabla h$. Here we present a reliability-aware two-stage computational workflow (`SubsurfaceReliability v1.0`) that integrates pre-training medium roughness screening, conservative face-flux operator learning, and online algebraic multigrid (AMG) residual screening and defect correction. Across 3,264 held-out evaluations spanning 24 lognormal heterogeneity regimes ($\sigma^2 \in [0.5, 4.0]$, $\lambda \in [0.02, 0.20]$), 12 unseen off-grid regimes, and non-Gaussian channelized and alluvial lens media, head-trained surrogates achieve median head errors of 5.9% (U-Net) and 5.5% (FNO) yet exceed the zero-velocity error threshold ($E_q > 1.0$) in 91.4% and 75.3% of evaluations. Spectral convolution analysis and Matérn covariance interventions demonstrate that wavenumber-squared (k^2) derivative weighting amplifies high-wavenumber head residuals across rough conductivity interfaces, while controlled decoder ablations show that bilinear resize-convolution eliminates U-Net's 2.26× finite-volume interface error amplification. In Stage 1, medium roughness invariants (σ^2/λ, σ^2, and discrete gradient energy $\mathcal{R}_{\nabla Y}$) screen regime reliability before training (ROC-AUC = 0.915–0.955; off-grid balanced accuracy 1.000 for U-Net and 100% sensitivity for FNO). Supervising FNO directly on two-point flux-approximation (TPFA) face fluxes cuts median velocity error by 52%–95% across Gaussian regimes and by 57%–73% zero-shot on channelized and alluvial lens media. In Stage 2, a single AMG V-cycle yields a reference-free residual indicator η_F (ROC-AUC = 0.997), cuts median face-flux error by 93% as an $O(N^2)$ defect correction, and raises Pollock advective particle breakthrough from 2.2% to 94.0% (median travel-time ratio 1.00).

Keywords: Neural PDE operators, Heterogeneous subsurface flow, Fourier neural operator,

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Physics-informed neural networks, Finite-volume flux conservation, Algebraic multigrid defect correction

1. Introduction

Numerical simulation of fluid flow and solute transport in heterogeneous porous formations underpins groundwater management, contaminant remediation, underground hydrogen storage, and geological carbon sequestration (Bear, 1972; Dagan, 1989; Gelhar, 1993; Blunt, 2017; Lie, 2019). Because hydraulic conductivity $K(\mathbf{x})$ varies by orders of magnitude across spatial scales (Delhomme, 1979; Rubin, 2003; Efendiev et al., 2013), ensemble data assimilation and uncertainty quantification require thousands of forward evaluations of the Darcy flow equation (Carrera et al., 2005; Laloy and Vrugt, 2012; Chen and Oliver, 2013; Emerick and Reynolds, 2013; Kitanidis and Lee, 2014). To accelerate these multi-query workflows, deep-learning surrogates and physics-informed neural solvers have been widely adopted across computational geoscience (Tripathy and Bilionis, 2018; Zhu and Zabaras, 2018; Raissi et al., 2019; Mo et al., 2019; Chang and Zhang, 2020; Karumuri et al., 2020; Tang et al., 2020; Li et al., 2021; Lu et al., 2021a; Karniadakis et al., 2021; Lauzon, 2024).

Three neural modeling paradigms predominate in subsurface flow computing: image-to-image convolutional encoder–decoders such as U-Net (Ronneberger et al., 2015; Zhu and Zabaras, 2018; Mo et al., 2019; Chang and Zhang, 2020; Wang et al., 2020; Lauzon, 2024), spectral neural operators such as the Fourier neural operator (FNO) (Li et al., 2021; Wen et al., 2022; Kovachki et al., 2023; Yan et al., 2025; Khazal et al., 2025), and instance-specific physics-informed neural networks (PINNs) (Sirignano and Spiliopoulos, 2018; E and Yu, 2018; Raissi et al., 2019; Tartakovsky et al., 2020; Bandai and Ghezzehei, 2021; Kashefi and Mukerji, 2022; Cuomo et al., 2022; Yan et al., 2024). In standard benchmarking protocols, amortized surrogates (U-Net and FNO) are trained by minimizing the mean-squared error (MSE) on hydraulic head h or pressure p, and model fidelity is reported via relative L_2 errors or R^2 on that state field (Zhu and Zabaras, 2018; Mo et al., 2019; Lu et al., 2022).

In subsurface hydrology and reservoir engineering, however, the primary quantity governing advective solute trajectories, capture-zone geometry, and residence times is the specific discharge (Darcy velocity) $\mathbf{q}(\mathbf{x}) = -K(\mathbf{x})\nabla h(\mathbf{x})$ (Raviart and Thomas, 1977; Pollock, 1988; Dagan, 1989; Rubin, 2003). Reconstructing $\mathbf{q}$ from a predicted head field $\hat{h}$ requires spatial differentiation followed by multiplication with a rough conductivity field $K(\mathbf{x}) = \exp(Y(\mathbf{x}))$. Although classical operator-norm and mixed finite-element analyses establish that L_2 convergence of a scalar potential does not bound its H^1 flux norm (Raviart and Thomas, 1977; Ern et al., 2010), the quantitative consequences of this decoupling in heterogeneous geological media remain poorly integrated into practical neural surrogate workflows. Recent studies have examined spectral bias in neural PDE solvers (Rahaman et al., 2019; Fuks and Tchelepi, 2020; Wang et al., 2021; Krishnapriyan et al., 2021; Wang et al., 2022; Grossmann et al., 2024), mixed pressure–velocity PINNs (Jagtap et al., 2020; Yu et al., 2022; Alhubail et al., 2024; Mamud et al., 2024), U-FNO architectures (Wen et al., 2022), and unsupervised finite-volume divergence losses for FNOs in mild heterogeneity ($\sigma^2 \leq 0.4$) (Yan et al., 2025). Nevertheless, three methodological gaps prevent reliable deployment across strong geological heterogeneity ($\sigma^2 \in [0.5, 4.0]$) and channelized or lens-dominated facies. First, practitioners

lack a pre-training screening rule that predicts, directly from $K(\mathbf{x})$ before training any network, when state-only supervision fails to yield usable velocity fields. Second, the relative efficacy of unsupervised finite-volume divergence penalties versus direct supervision on harmonic face fluxes has not been isolated against matched-budget optimization controls in rough media. Third, even when derivative-aware training reduces global L_2 velocity error below unity, residual cell-level divergence errors can still sever advective streamlines, necessitating a reference-free online diagnostic and fast conservative correction.

Here we develop a unified, reliability-aware two-stage computational workflow (`SubsurfaceReliability v1.0`) that resolves these gaps by linking spectral operator-norm diagnosis, pre-training medium roughness screening (Stage 1), conservative two-point flux-approximation (TPFA) face-flux operator training, and online single-cycle algebraic multigrid (AMG) residual screening and defect correction (Stage 2). We formalize the regime of *flux blindness*, in which a solver achieves low relative L_2 head error ($E_h \approx 5\%$–6%) while its derived Darcy velocity field exceeds unit relative error ($E_q > 1.0$, worse than predicting zero flow $\hat{\mathbf{q}} = \mathbf{0}$). Using a verified benchmark suite spanning 24 lognormal regimes (3,264 evaluations), 12 prospective off-grid blind regimes, and non-Gaussian channelized and Macrodispersion Experiment (MADE)-type alluvial lens media, we establish the spectral mechanism of state–velocity decoupling, validate transferable pre-training screening rules, demonstrate why direct TPFA face-flux supervision outperforms unsupervised divergence penalties, and quantify the accuracy–cost trade-offs of hybrid neural–AMG defect correction for advective transport.

2. Computational Workflow and Experimental Design

Our computational framework (`SubsurfaceReliability v1.0`) is structured around the four-stage verification and evaluation pipeline summarized in Table 1. Full hyperparameter sensitivity checks, vector-norm comparisons, and multi-seed tables are provided in the Supplementary Material.

2.1. Governing Equations and Spectral Medium Synthesis

We consider steady-state, single-phase Darcy flow on the dimensionless unit square $\Omega = [0, L]^2$ ($L = 1$):

$$-\nabla \cdot \big(K(\mathbf{x})\nabla h(\mathbf{x})\big) = 0, \quad \mathbf{x} = (x, y) \in \Omega, \tag{1}$$

where $h(\mathbf{x})$ is dimensionless hydraulic head and $\mathbf{q}(\mathbf{x}) = -K(\mathbf{x})\nabla h(\mathbf{x})$ is the Darcy velocity, subject to Dirichlet inflow/outflow and lateral no-flow boundary conditions:

$$h(0, y) = 1, \quad h(1, y) = 0, \quad \frac{\partial h}{\partial y}(x, 0) = \frac{\partial h}{\partial y}(x, 1) = 0. \tag{2}$$

In the primary 24-regime benchmark atlas, hydraulic conductivity is modeled as a lognormal field $K(\mathbf{x}) = \exp(Y(\mathbf{x}))$ (Dagan, 1989; Gelhar, 1993; Dietrich and Newsam, 1993), where $Y(\mathbf{x}) = \ln K(\mathbf{x})$ is a zero-mean Gaussian field with variance σ^2, dimensionless correlation length λ, and isotropic Gaussian autocovariance:

$$C_Y(r) = \sigma^2 \exp\left(-\frac{r^2}{2\lambda^2}\right), \quad r = \|\mathbf{x} - \mathbf{x}'\|_2, \tag{3}$$

Table 1: Summary of solver architectures, spatial representations, training protocols, and evaluation metrics in the `SubsurfaceReliability v1.0` workflow.

Component	**TPFA Reference**	**Convolutional U-Net**	**Fourier Operator (FNO)**	**Coordinate PINN**
Paradigm	Direct FV solver	Amortized 2D CNN	Amortized spectral net	Per-instance MLP
Input tensor	Cell $K_{i,j}$ $(128^2, 256^2)$	$[Y_{\text{std}}, x, y] \in \mathbb{R}^{3\times N\times N}$	$[Y_{\text{std}}, x, y] \in \mathbb{R}^{3\times N\times N}$	$(x, y) \in [0,1]^2$; $4N$ $(Y, \nabla Y)$
Architecture	Harmonic face $T_{i+1/2,j}$	$w = 24$, $d = 3$, GroupNorm	$w = 24$, $L = 4$, $m = 12$ modes	4×64 tanh; hard BC
Parameters	None ($A_K\mathbf{h} = \mathbf{b}$)	263,689 (Resize: 292,489)	669,377 tensor elements	12,737 per realization
Objective	Exact $\nabla_h \cdot \mathbf{F} = 0$	$\mathcal{L}_h$ or $\mathcal{L}_h + (\alpha/f_0)\mathcal{L}_q$	$\mathcal{L}_h$ or $\mathcal{L}_h + (\alpha/f_0)\mathcal{L}_q$	PDE residual (r_{norm} or Kr)
Optimization	Direct LU (41 ms)	Adam (80 ep, lr 10^{-3})	Adam (80 ep, lr 10^{-3})	10,000 Adam + 500 L-BFGS
Data split/cell	352 realizations	128 train / 32 val / 64 test	128 train / 32 val / 64 test	8 stratified test instances
Evaluations	MMS + 256^2 check	1,536 atlas + 832 o.o.d.	1,536 atlas + 832 o.o.d.	192 atlas + 8 blind
Diagnostics	Reference ($E_q = 0$)	$E_h, E_q, E_{q,\text{face}}, \mathcal{I}_{\text{div}}, \eta_F$	$E_h, E_q, E_{q,\text{face}}, \mathcal{I}_{\text{div}}, \eta_F$	$E_h, E_q, E_{q,\text{face}}, \mathcal{I}_{\text{div}}, \eta_F$

having analytical radial integral scale $I_Y = \lambda\sqrt{\pi/2}$. To ensure exact physical equivalence between grid-based operators and continuous coordinate networks, each realization $Y(\mathbf{x})$ is synthesized via its exact band-limited trigonometric Fourier expansion with zero mean and exact sample variance σ^2. Grid solvers evaluate $Y(\mathbf{x})$ at the $N \times N$ cell centers ($N = 128$, and $N = 256$ in the resolution-scaling benchmark), whereas the PINN queries bilinear interpolants of exact spectral $(Y, \partial_x Y, \partial_y Y)$ fields precomputed on a $4N \times 4N$ grid (interpolation error 4×10^{-7}).

To isolate small-scale roughness at fixed variance σ^2 and fixed integral scale $I_Y = \lambda\sqrt{\pi/2}$, we also construct controlled Matérn covariance interventions with smoothness $\nu = 3/2$ and $\nu = 1/2$ (Rubin, 2003):

$$C_{\nu=3/2}(r) = \sigma^2\left(1 + \frac{r}{\ell_{3/2}}\right)\exp\left(-\frac{r}{\ell_{3/2}}\right), \quad C_{\nu=1/2}(r) = \sigma^2\exp\left(-\frac{r}{\ell_{1/2}}\right), \tag{4}$$

with $\ell_{3/2} = I_Y/2$ and $\ell_{1/2} = I_Y$, quantifying high-wavenumber energy via the spectral fraction $\Phi(k_c = 8) = E_Y(k \geq 8)/E_{Y,\text{tot}}$. Beyond multi-Gaussian fields, we evaluate two non-Gaussian hydrofacies ensembles on 128×128 grids (16 realizations per regime): (i) sinuous binary-facies channelized aquifers (35% connected high-K channels across $\sigma^2 \in \{0.5, 1.0, 2.0\}$) and (ii) strongly heterogeneous three-facies alluvial cross-sections ($\sigma^2 = 4.5$, gravel conduits, sand matrix, and silt-clay lenses with anisotropy $\lambda_x = 0.22, \lambda_z = 0.03$) modeled after the Macrodispersion Experiment (MADE) aquifer (Zheng et al., 2011).

2.2. Verified Finite-Volume Reference Solver and Atlas Design

Reference solutions are computed using a locally conservative cell-centered TPFA finite-volume scheme (Eymard et al., 2000). Across adjacent cells, the normal face flux is governed by harmonic transmissibility weighting:

$$F_{i+1/2,j} = -T_{i+1/2,j}\,\frac{h_{i+1,j} - h_{i,j}}{\Delta x}, \quad T_{i+1/2,j} = \frac{2K_{i,j}K_{i+1,j}}{K_{i,j} + K_{i+1,j}}, \tag{5}$$

yielding a symmetric positive-definite system $A_K\mathbf{h} = \mathbf{b}$ solved via sparse direct factorization. Automated verification gates (`src/subrel/verify.py`) confirm second-order convergence (2.00 in head, 1.98 in flux) under the method of manufactured solutions (Roache, 1998), exact linear head recovery (1.0×10^{-14}), and machine-precision cell divergence closure ($\leq 8.6 \times 10^{-12}$). Grid-refinement checks between 128^2 and 256^2 grids confirm that discretization differences (0.03%–6.1% across the Gaussian atlas) are an order of magnitude smaller than neural flux errors (Supplementary Section S1).

The full-factorial atlas spans $\sigma^2 \in \{0.5, 1.0, 2.0, 4.0\} \times \lambda \in \{0.02, 0.03, 0.05, 0.08, 0.12, 0.20\}$ (24 regimes, each with 128 baseline training, 128 reserve training, 32 validation, and 64 held-out test realizations). Evaluating U-Net and FNO across all 64 test realizations per cell (1,536 evaluations each) and per-instance PINNs across the first 8 test realizations per cell (192 evaluations) yields 3,264 primary atlas evaluations, complemented by 12 off-grid blind regimes (768 test realizations).

2.3. Neural Solver Architectures and Training Protocols

All baseline solvers use uniform hyperparameters across regimes (Table 1). The baseline 2D U-Net (Ronneberger et al., 2015) maps $[Y_{\text{std}}(\mathbf{x}), x, y] \in \mathbb{R}^{3\times N\times N}$ to normalized head $\tilde{h} = 2\hat{h} - 1$ using base width $w = 24$, three resolution levels ($24 \to 48 \to 96$), 2×2 average pooling, and stride-2 transposed-convolution upsampling (263,689 parameters). To isolate decoder-induced interface oscillations, we also evaluate a controlled variant (`ResizeUNet`, 292,489 parameters) that replaces stride-2 transposed convolutions with bilinear upsampling followed by a 3×3 convolution. The 2D FNO (Li et al., 2021) lifts $[Y_{\text{std}}(\mathbf{x}), x, y]$ to width $w = 24$ followed by $L = 4$ spectral convolution layers retaining $m_1 = m_2 = 12$ Fourier modes (669,377 parameter elements). Both amortized families are trained on 128 realizations for 80 epochs (batch size 32, Adam learning rate 10^{-3} with cosine annealing) using validation early stopping.

The per-instance coordinate PINN (Raissi et al., 2019) (4×64 tanh layers, 12,737 parameters) enforces the Dirichlet conditions exactly via the hard trial ansatz (Lu et al., 2021c):

$$\hat{h}(x, y; \theta) = (1 - x) + x(1 - x)\,\mathcal{N}_\theta(x, y), \tag{6}$$

and minimizes the conductivity-normalized residual:

$$r(x, y; \theta) = \Delta\hat{h}(x, y; \theta) + \nabla Y(x, y) \cdot \nabla\hat{h}(x, y; \theta), \tag{7}$$

plus a lateral Neumann penalty over $M = 16{,}384$ Latin-hypercube interior collocation points and $M_{\text{bc}} = 1{,}024$ boundary points for 10,000 Adam iterations followed by 500 L-BFGS steps. We also evaluate residual-adaptive resampling (RAR) (Lu et al., 2021b) and conductivity-weighted residuals $K(\Delta\hat{h} + \nabla Y \cdot \nabla\hat{h})$ in Section 3.4.

2.4. Evaluation Metrics and Streamline Transport Protocol

All error metrics are evaluated on the interior domain Ω_{int} excluding a 2-cell boundary margin. The relative L_2 hydraulic-head error and Darcy velocity error are defined as

$$E_h = \frac{\|\hat{h} - h\|_{2,\Omega_{\text{int}}}}{\|h\|_{2,\Omega_{\text{int}}}}, \tag{8}$$

$$E_q = \frac{\|\hat{q}_x - q_x\|_{2,\Omega_{\text{int}}} + \|\hat{q}_y - q_y\|_{2,\Omega_{\text{int}}}}{\|q_x\|_{2,\Omega_{\text{int}}} + \|q_y\|_{2,\Omega_{\text{int}}}}, \tag{9}$$

and the conservative TPFA face-flux error $E_{q,\text{face}}$ is defined analogously on the interior cell faces using Eq. (5). A velocity prediction is classified as a *flux failure* when $E_q > 1.0$ (worse than predicting $\hat{\mathbf{q}} = \mathbf{0}$); replacing Eq. (9) with the joint Euclidean vector norm alters only 5.2% of binary classifications across all 3,072 amortized evaluations and leaves all screening rules unchanged (Supplementary Section S2). Local mass-balance violation is quantified by the dimensionless divergence imbalance:

$$\mathcal{I}_{\text{div}} = \frac{\|\nabla_h \cdot \hat{\mathbf{q}}_f\|_{2,\Omega_{\text{int}}}}{(\|q_{f,x}\|_{2,\Omega_{\text{int}}} + \|q_{f,y}\|_{2,\Omega_{\text{int}}})/\Delta x}. \tag{10}$$

Downstream advective solute transport is evaluated via semi-analytical Pollock streamline tracking (Pollock, 1988) on the reconstructed TPFA face fluxes across 24 test realizations (41 particles released at $x = 2\Delta x$ per realization, 984 trajectories per solver) spanning $(\sigma^2, \lambda) \in \{(0.5, 0.08), (1.0, 0.05), (2.0, 0.03)\}$, recording whether each trajectory arrives at the outlet ($x = 1 - 2\Delta x$), stalls in an artificial interior sink cell, or exits backward through the inflow boundary.

3. Results

3.1. The Empirical Reliability Atlas and State–Velocity Decoupling

Across all 24 heterogeneity regimes (3,264 held-out evaluations), head-trained amortized surrogates achieve high accuracy on the state variable h, with pooled median head errors of 5.9% for U-Net and 5.5% for FNO, whereas the per-instance PINN exhibits a larger median head error of 27.1% due to optimization stiffness in rough media (Table 2; Figure 1). When evaluated on the derived Darcy velocity $\mathbf{q} = -K\nabla h$, however, the performance ranking reverses. Despite head errors below 6%, the baseline U-Net yields a pooled median flux error of $E_q = 2.75$, exceeding the zero-velocity threshold $E_q = 1.0$ in 91.4% of realizations and achieving regime-median skill ($E_q \leq 1.0$) in only 1/24 regimes ($\sigma^2 = 0.5, \lambda = 0.20$). The baseline FNO yields a pooled median flux error of $E_q = 1.71$ (75.3% failure rate), maintaining regime-median skill in 7/24 regimes (all six $\sigma^2 = 0.5$ cells plus $(1.0, 0.20)$). By contrast, the per-instance PINN achieves a pooled median flux error of 0.90 and satisfies $E_q \leq 1.0$ in 14/24 regimes.

Evaluating velocity accuracy on conservative TPFA face fluxes exposes a pronounced interface penalty in the baseline U-Net: its 24-regime median TPFA face-flux error reaches 5.48 (2.26× its cell-centered median of 2.42), producing a large divergence imbalance of $\mathcal{I}_{\text{div}} = 8.27$ because $2\Delta x$ central differences cancel period-$2\Delta x$ transposed-convolution oscillations

Table 2: Atlas-wide performance summary across 24 lognormal heterogeneity regimes (3,264 held-out evaluations). Note on computational budget: U-Net and FNO are amortized surrogates trained once per regime (95–110 s) and evaluated in a single forward pass (2.8 ms/realization) across 64 test instances/cell, whereas PINN is an instance-specific PDE optimizer (10,500 steps, $\sim$ 95 s/realization) evaluated on 8 stratified test instances/cell. Cell-cluster bootstrap 95% confidence intervals (1,000 resamples) are reported for each family's governing pre-training ROC-AUC predictor.

Solver Family (Paradigm, Test N)	Med. E_h	95th E_h	Med. E_q	Fail Rate	Med. $E_{q,\text{face}}$	Med. $\mathcal{I}_{\text{div}}$	Pre-Train AUC [95% CI]
U-Net (amortized, 1,536, 1/24 pass)	5.9%	15.7%	2.75	91.4%	5.48	8.27	σ^2/λ: 0.955 [0.887, 0.980]
FNO (amortized, 1,536, 7/24 pass)	5.5%	15.2%	1.71	75.3%	1.67	1.01	σ^2: 0.915 [0.844, 0.953]
PINN (per-instance, 192, 14/24 pass)	27.1%	65.5%	0.90	44.3%	0.98	0.03	σ^2/λ^2: 0.938 [0.865, 0.979]

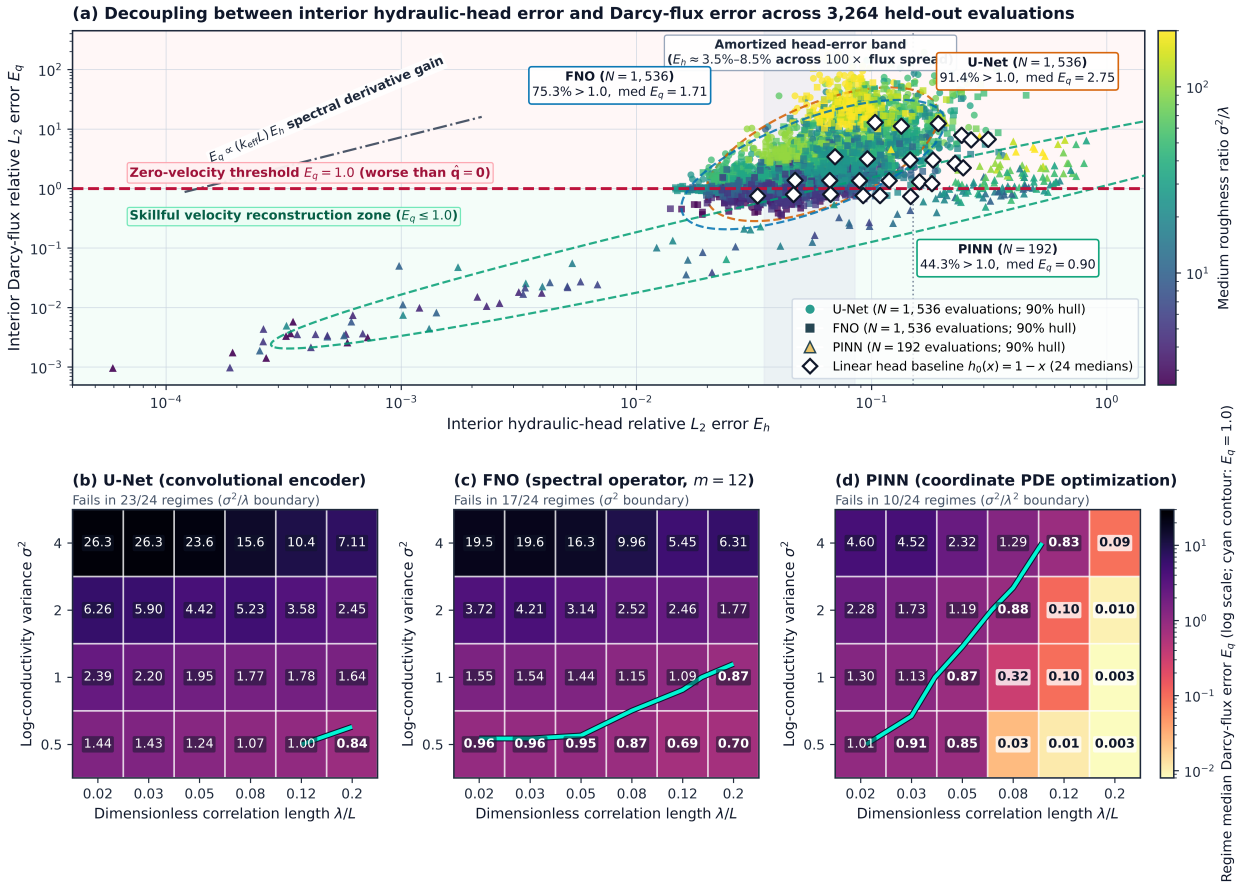


Figure 1: **The subsurface flow reliability atlas across 24 heterogeneity regimes (3,264 held-out evaluations). Top (a):** Instance-level hydraulic-head relative L_2 error (E_h) versus interior Darcy-flux relative L_2 error (E_q) on logarithmic axes, colored by the dimensionless roughness ratio σ^2/λ. Dashed ellipses show 90% covariance hulls, and open white diamonds denote the untrained linear head baseline $h_0(x) = 1 - x$, which achieves lower Darcy-flux error than head-trained U-Net and FNO models across the entire short-correlation half of the atlas ($\lambda \leq 0.05$). **Bottom (b–d):** Regime-median interior Darcy-flux error E_q across the (σ^2, λ) grid for U-Net, FNO, and PINN. The cyan contour traces the skill boundary $E_q = 1.0$.

whereas single-cell TPFA face gradients differentiate directly across them. By comparison, FNO's regime-median face-flux error is 1.67 ($\mathcal{I}_{\text{div}} = 1.01$), and PINN achieves 0.98 ($\mathcal{I}_{\text{div}} = 0.03$). Strikingly, across the entire short-correlation half of the atlas ($\lambda \leq 0.05$), the zero-

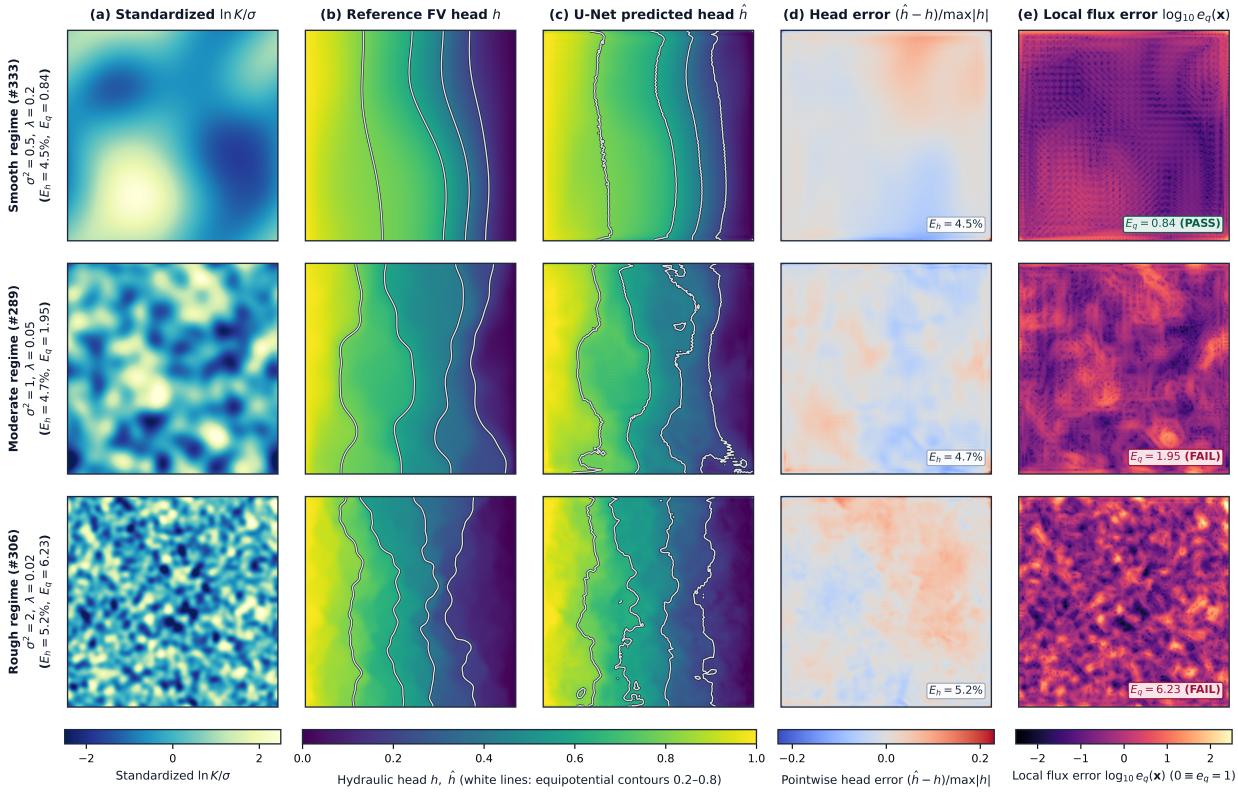


Figure 2: **Spatial anatomy of flux blindness for the U-Net surrogate across smooth ($\sigma^2 = 0.5, \lambda = 0.20$), moderate ($\sigma^2 = 1.0, \lambda = 0.05$), and rough ($\sigma^2 = 2.0, \lambda = 0.02$) regimes.** Columns display (a) standardized log-conductivity $\ln K/\sigma$, (b) finite-volume reference head h, (c) U-Net predicted head $\hat{h}$, (d) normalized pointwise head error $(\hat{h}-h)/\max|h|$, and (e) logarithmic Darcy-flux error magnitude $\log_{10} e_q(\mathbf{x})$.

training linear profile $h_0(x) = 1 - x$ achieves lower Darcy-flux error than both head-trained U-Net and FNO surrogates (Figure 1a). Figure 2 illustrates this spatial anatomy for U-Net: while predicted head contours closely match the reference ($E_h = 4.5\%$–5.2%), grid-scale head residuals concentrated along high-∇Y interfaces generate severe velocity errors ($E_q = 1.95$ and 6.23).

3.2. Spectral Amplification Mechanism and Covariance Interventions

Writing $\delta h = \hat{h} - h$ for the head error and $\delta\mathbf{q} = \hat{\mathbf{q}} - \mathbf{q}$ for the Darcy velocity error under the exact conductivity field $K(\mathbf{x}) = \exp(Y(\mathbf{x}))$ yields the exact operator relation

$$\delta\mathbf{q}(\mathbf{x}) = -K(\mathbf{x})\nabla\delta h(\mathbf{x}), \tag{11}$$

which transforms in two-dimensional Fourier space into the spectral convolution identity

$$\widehat{\delta\mathbf{q}}(\mathbf{k}) = -\sum_{\mathbf{k}'} \widehat{K}(\mathbf{k}-\mathbf{k}')\, i\mathbf{k}'\, \widehat{\delta h}(\mathbf{k}'). \tag{12}$$

Equation (12) shows that spatial differentiation weights each head-error Fourier mode $\widehat{\delta h}(\mathbf{k}')$ by wavenumber $k' = |\mathbf{k}'|$ (k'^2 in energy) while convolution with $\widehat{K}$ couples errors across

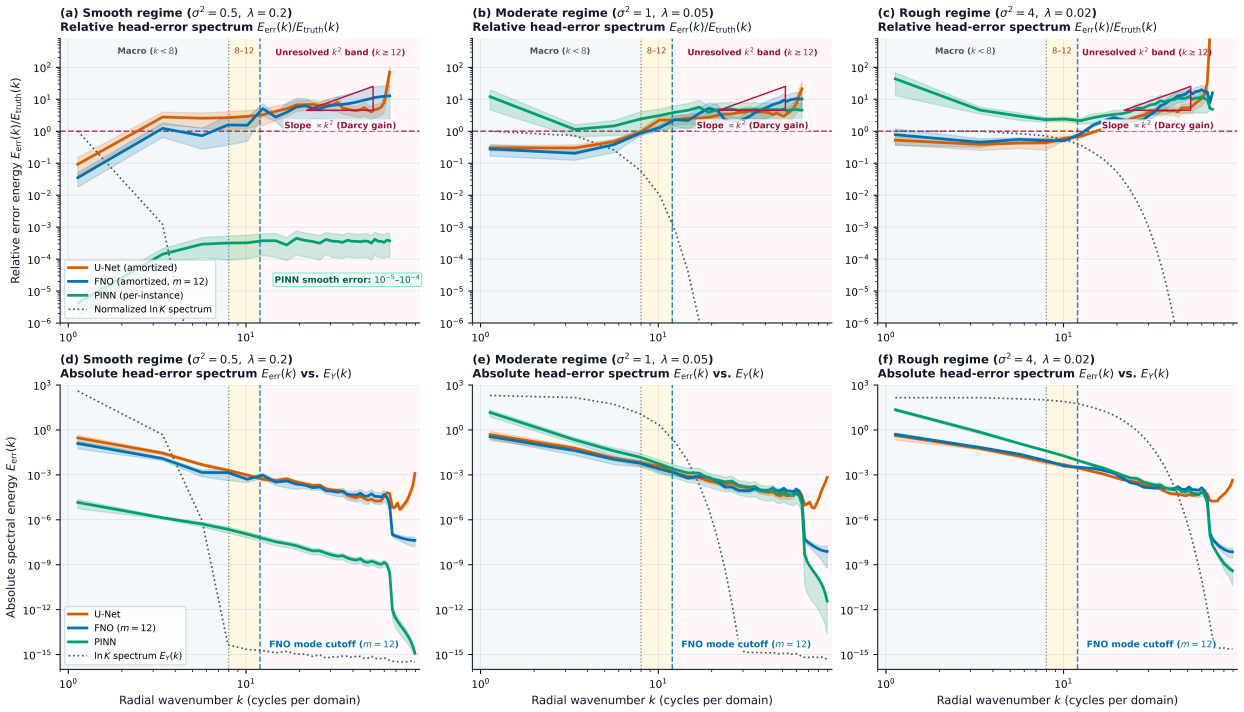


Figure 3: **Radially averaged spectral analysis across smooth** ($\sigma^2 = 0.5, \lambda = 0.20$)**, moderate** ($\sigma^2 = 1.0, \lambda = 0.05$)**, and rough** ($\sigma^2 = 4.0, \lambda = 0.02$) **regimes. Top (a–c):** Head-error spectral energy normalized by the reference head-fluctuation spectrum (± 1 SD ribbons), with red triangles indicating the k^2 derivative-gain slope. **Bottom (d–f):** Absolute head-error power spectra compared against the log-conductivity spectrum $E_Y(k)$ (dotted curve) and the FNO mode cutoff ($m = 12$).

bandwidth $\propto 1/\lambda$. Radially averaged error spectra across smooth $(0.5, 0.20)$, moderate $(1.0, 0.05)$, and rough $(4.0, 0.02)$ regimes confirm this mechanism (Figure 3): relative to the reference head-fluctuation spectrum, U-Net and FNO head-error energies grow along the analytical k^2 slope and exceed the true signal energy by one to two orders of magnitude at $k \geq 16$. For U-Net, transposed-convolution upsampling produces a flat high-wavenumber floor across $k \in [12, 64]$; for FNO, mode truncation at $m = 12$ leaves modes $k > 12$ unattenuated; whereas PINN's error spectrum decays two to four orders of magnitude lower across $k \in [6, 64]$.

Controlled covariance-family interventions across Gaussian, Matérn $\nu = 3/2$, and Matérn $\nu = 1/2$ media at matched variance σ^2 and matched integral scale I_Y confirm that small-scale roughness drives velocity error independently of (σ^2, λ) (Figure 4). As the high-wavenumber energy fraction $\Phi(k \geq 8)$ increases from Gaussian to Matérn $\nu = 1/2$, median Darcy-flux error rises monotonically by $+53\%$ $(1.95 \to 2.99)$ at $(1.0, 0.05)$ and $+16\%$ $(5.90 \to 6.83)$ at $(2.0, 0.03)$ for U-Net, and by $+20\%$ $(1.03 \to 1.24)$ and $+72\%$ $(1.91 \to 3.29)$ for PINN. FNO remains nearly invariant across covariance tails $(1.44 \to 1.57$ and $4.21 \to 3.96)$ because its $m = 12$ spectral truncation makes baseline FNO error primarily variance-governed. Phase-preserving Fourier shell redistribution at $(1.0, 0.05)$ and capacity ablations (expanding FNO to $m = 32$ modes or tripling U-Net parameters to 730,041; Supplementary Section S3) confirm that scaling capacity under a head-only loss yields modest 4.5%–31% refinements

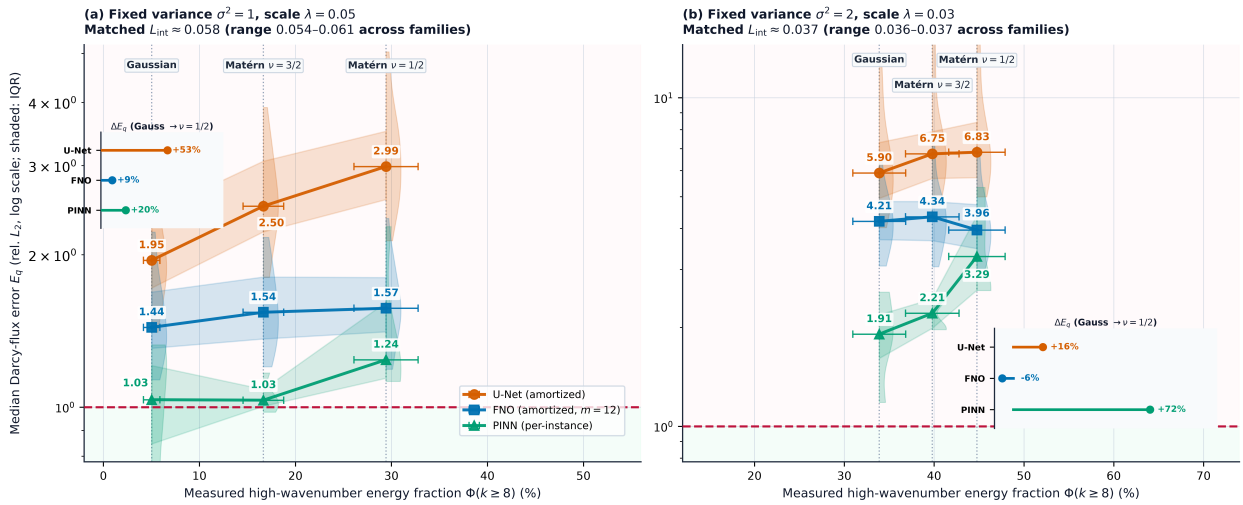


Figure 4: **Controlled covariance-family intervention at matched variance and integral scale.** Median interior Darcy-flux error E_q, 25%–75% interquartile ribbons, and marginal density distributions plotted against the high-wavenumber spectral energy fraction $\Phi(k \geq 8)$ across Gaussian, Matérn $\nu = 3/2$, and Matérn $\nu = 1/2$ media at $(\sigma^2, \lambda) = (1.0, 0.05)$ and $(2.0, 0.03)$.

without shifting the regime failure boundary.

3.3. Stage 1: Pre-Training Roughness Screening Across Gaussian and Non-Gaussian Media

Evaluating normalized head-error fields $\delta\mathbf{h}/\|\delta\mathbf{h}\|_2$ reveals that pointwise head errors are spatially phase-locked to the specific high-∇Y interfaces of each realization, rendering error fields nearly orthogonal across realizations (mean cosine similarity 0.04–0.08; Supplementary Section S4). Consequently, transferable pre-training predictability resides in scalar roughness invariants computed directly from the conductivity ensemble $K(\mathbf{x})$. Once calibrated on the baseline benchmark archive, evaluating these descriptors on a new geological ensemble requires zero neural training or PDE solves.

Across all 3,264 atlas evaluations, architecture-specific medium invariants separate reliable from failing evaluations with instance-level ROC-AUCs of 0.955 for U-Net (σ^2/λ), 0.915 for FNO (σ^2), and 0.938 for PINN (σ^2/λ^2) (Table 3). When variogram parameters (σ^2, λ) are unknown, computing the parameter-free discrete gradient energy $\mathcal{R}_{\nabla Y} = L^2\langle|\nabla \ln K|^2\rangle$ directly on the grid yields nearly identical instance-level ROC-AUCs of 0.937 (U-Net), 0.797 (FNO), and 0.935 (PINN). Because the baseline U-Net fails in 23/24 atlas regimes and 11/12 blind regimes, raw regime accuracy is confounded by class prevalence; Table 3 therefore reports sensitivity (Sens.), specificity (Spec.), false negative rate (FNR), false positive rate (FPR), and balanced accuracy ($\frac{1}{2}$(Sens. + Spec.), where a majority-class classifier scores 0.500). Under nested leave-one-regime-out (LOO) cross-validation using Youden's J statistic, the governing descriptors achieve instance-level balanced accuracies of 0.910 (U-Net), 0.853 (FNO), and 0.812 (PINN).

On the 12 unseen off-grid regimes spanning interior interpolations, transition cells, and variance extrapolations up to $\sigma^2 = 6.0$ (Supplementary Table S1), the frozen U-Net rule ($\sigma^2/\lambda \geq 3.33$) classifies all 12/12 regimes correctly (balanced accuracy = 1.000, FNR = 0.00, FPR = 0.00), identifying the single reliable regime $(0.75, 0.26)$ ($E_q = 0.89$) and flagging

Table 3: Stage 1 pre-training roughness screening across the 24-regime atlas and 12 prospective off-grid blind regimes (64 test realizations/regime). To account for class prevalence in baseline U-Net (23/24 atlas and 11/12 blind regimes failing), threshold-free instance ROC-AUC, sensitivity (Sens.), specificity (Spec.), false negative rate (FNR), false positive rate (FPR), and balanced accuracy (Bal. Acc., versus 0.500 for a majority-class guess) are reported. Thresholds ($\sigma^2/\lambda \geq 3.33$ for U-Net, $\sigma^2 \geq 0.75$ for FNO) were frozen prior to blind testing.

Solver	Descriptor	Instance AUC [95% CI]	LOO Bal. Acc. (Sens./Spec.)	Regime LOO	Blind Raw	Blind Bal. Acc. (Sens./Spec./FNR/FPR)
U-Net	σ^2/λ	0.955 [0.887, 0.980]	0.910 (0.90/0.92)	23/23 (maj. 23/23)	12/12	1.000 (1.00/1.00/0.00/0.00)
FNO	σ^2	0.915 [0.844, 0.953]	0.853 (0.86/0.85)	23/24 (maj. 17/24)	8/12	0.688 (1.00/0.38/0.00/0.62)
PINN	σ^2/λ^2 ($\mathcal{R}_{\nabla Y}$)	0.938 [0.865, 0.979]	0.812 (0.82/0.80)	22/24 (maj. 14/24)	2/2	1.000 (1.00/1.00/0.00/0.00)

extreme amplification at $(6.0, 0.065)$ ($E_q = 82.3$). The frozen FNO rule ($\sigma^2 \geq 0.75$) acts as a zero-false-negative safety screen (sensitivity = 1.000, FNR = 0.00 across all 8 failing regimes), conservatively flagging borderline $\sigma^2 = 0.75$ regimes where flux-informed training is warranted. While the dimensionless invariants ($\sigma^2/\lambda, \sigma^2, \mathcal{R}_{\nabla Y}$) reflect operator structure, the numerical cutoffs are specific to the baseline architecture and training budget.

To test generalizability beyond multi-Gaussian fields, we applied the parameter-free gradient screen $\mathcal{R}_{\nabla Y}$ to the three sinuous binary-facies channelized regimes ($\sigma^2 \in \{0.5, 1.0, 2.0\}$, 16 realizations/regime, $n = 144$ model evaluations) and the MADE-type heterogeneous alluvial lens cross-sections ($\sigma^2 = 4.5$, 16 realizations, $n = 48$ evaluations; `src/subrel/extended_controls.py`). Without variogram fitting, $\mathcal{R}_{\nabla Y}$ yields 354.5, 1,141.4, and 3,673.2 across the three channelized regimes and 6,061.2 on the MADE alluvial cross-sections, flagging sharp facies boundaries. Consistent with these high roughness values, head-trained U-Net and head-trained FNO fail in velocity ($E_q > 1.0$) across all three channelized regimes ($E_q = 1.32$–1.43 and 1.07–1.44) and on the MADE cross-sections ($E_{q,\text{face}} = 3.34$ and 1.76).

3.4. Flux-Informed Operator Learning vs. Unsupervised Divergence Penalties

Before evaluating direct face-flux supervision, we benchmarked standard training modifications against matched-budget controls (Figure 5). While doubling the training set (N_{train} : $128 \to 256$) stabilizes mild-variance boundary cells such as $(0.5, 0.02)$ (E_q : $1.44 \to 0.95$ for U-Net and $0.96 \to 0.54$ for FNO) and conductivity-weighted PDE residuals cut rough-regime PINN flux error by 53% at $(2.0, 0.03)$ ($1.91 \to 0.91$), unsupervised finite-volume divergence penalties exhibit a critical optimization bottleneck in rough media. Specifically, across six threshold-adjacent regimes, we compared 30-epoch fine-tuning with an unsupervised TPFA divergence penalty ($\|\nabla_h \cdot \hat{\mathbf{q}}_f\|_2^2$, $\beta = 0.3$, analogous to recent finite-volume physics-informed neural operator losses (Yan et al., 2025)) against a matched-budget control continuing plain head-MSE training ($\beta = 0$) for the same 30 epochs (Figure 5a). Plain MSE continuation achieves equal or lower velocity error on four of the six regimes. Mathematically, although $-\nabla \cdot (K \nabla h) = 0$ uniquely determines h in the continuum limit, the discrete TPFA divergence operator $\nabla_h \cdot (T \nabla_h \hat{h})$ applies a second-order difference operator to $\hat{h}$ (whose squared L_2 penalty scales as $O(k^4)$ for smooth K on uniform grids and becomes severely ill-conditioned across discontinuous K interfaces), creating strong gradient stiffness during stochastic optimization.

By contrast, supervising the surrogate directly on reference conservative TPFA face fluxes penalizes a first-order difference in $\hat{h}$ ($O(k^2)$ energy-norm conditioning) at zero additional

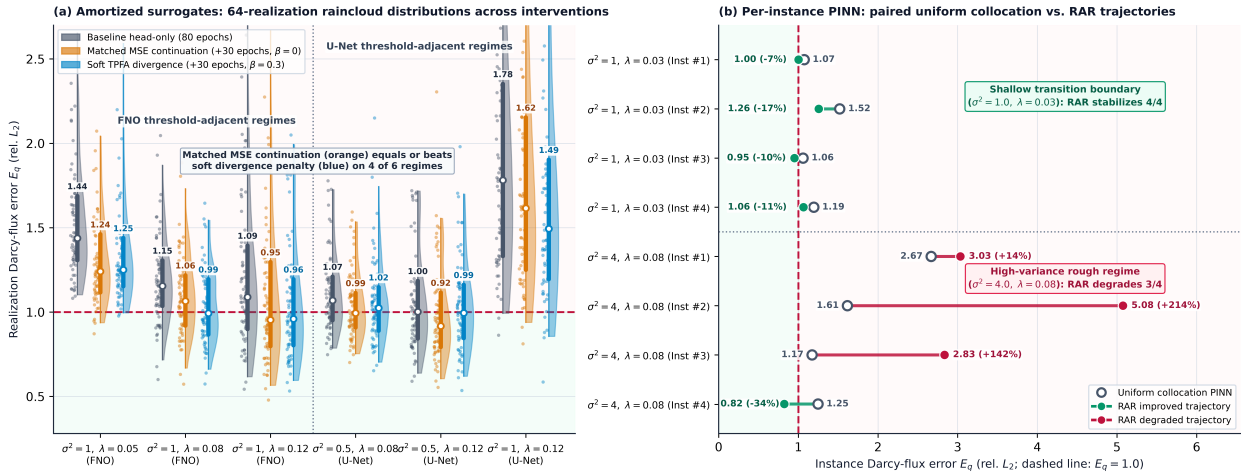


Figure 5: **Controlled evaluation of training modifications. Left (a):** Across six threshold-adjacent regimes, 64-realization raincloud distributions compare the baseline model (80 epochs), a matched-budget MSE continuation control (+30 epochs, $\beta = 0$), and unsupervised TPFA divergence fine-tuning (+30 epochs, $\beta = 0.3$). **Right (b):** Per-instance PINN flux-error trajectories from uniform collocation to residual-adaptive resampling (RAR) at $(1.0, 0.03)$ and $(4.0, 0.08)$.

Table 4: Effect of conservative TPFA face-flux supervision (Eq. (13)) on regime-median interior Darcy-flux error E_q across seven Gaussian regimes ($n = 64$ test realizations/regime) and zero-shot generalization to non-Gaussian sinuous channelized ($n = 16$ realizations/regime) and MADE-type alluvial lens ($n = 16$ realizations) media. Bold values satisfy $E_q \leq 1.0$.

Regime (σ^2, λ) **or Facies Type**	**FNO** $\alpha = 0$	**FNO** $\alpha = 0.01$	**FNO** $\alpha = 0.10$	**U-Net** $\alpha = 0$	**U-Net** $\alpha = 0.01$	**U-Net** $\alpha = 0.10$
$(0.5, 0.20)$ Smooth Gaussian	**0.70**	**0.32** (−53%)	**0.20** (−71%)	**0.84**	**0.79** (−6%)	1.04 (+24%)
$(0.5, 0.08)$ Mild Gaussian	**0.87**	**0.31** (−65%)	**0.19** (−78%)	1.07	1.01 (−6%)	1.07 (0%)
$(1.0, 0.12)$ Moderate λ	1.09	**0.47** (−56%)	**0.30** (−73%)	1.78	1.71 (−4%)	1.94 (+9%)
$(1.0, 0.05)$ Transition Gaussian	1.44	**0.51** (−64%)	**0.38** (−73%)	1.95	1.88 (−3%)	2.12 (+9%)
$(2.0, 0.08)$ Moderate σ^2	2.52	1.20 (−52%)	**0.93** (−63%)	5.23	5.27 (+1%)	4.57 (−13%)
$(2.0, 0.03)$ Rough Gaussian	4.21	1.45 (−65%)	1.03 (−76%)	5.90	5.53 (−6%)	4.74 (−20%)
$(4.0, 0.02)$ Extreme Gaussian	19.55	5.24 (−73%)	1.04 (−95%)	26.34	25.14 (−5%)	20.56 (−22%)
Sinuous Channelized ($\sigma^2 = 2.0$, $n = 16$)	1.44	**0.42** (−71%)	—	1.43	—	—
MADE Alluvial Lenses ($\sigma^2 = 4.5$, $n = 16$)	1.75	**0.75** (−57%)	—	1.93	—	—

simulation cost:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_h + \frac{\alpha}{f_0} \frac{\|\mathbf{q}_f(\hat{h}, K) - \mathbf{q}_f(h, K)\|^2_{2,\Omega_{\text{int}}}}{\|\mathbf{q}_f(h, K)\|^2_{2,\Omega_{\text{int}}}}, \tag{13}$$

where $\alpha \in \{0.01, 0.10\}$ is normalized by the initial epoch-0 face-flux loss f_0. Across seven benchmark regimes (28 trained models under identical seeds and epoch budgets; Table 4 and Figure 6), setting $\alpha = 0.01$ cuts FNO median Darcy-flux error by 52%–73%, while $\alpha = 0.10$ cuts FNO median flux error by 63%–95% (moving five of seven regimes below $E_q = 1.0$ and reducing $(2.0, 0.03)$ and $(4.0, 0.02)$ from 4.21 and 19.55 to 1.03 and 1.04). Crucially, evaluating the $\alpha = 0.01$ Flux-FNO zero-shot on non-Gaussian media cuts median face-flux error by 68%–73% across all three sinuous channelized regimes ($E_{q,\text{face}} = 0.35, 0.34, 0.40 \leq 1.0$) and by 58% on the MADE-type alluvial lens cross-sections ($E_{q,\text{face}} : 1.76 \to 0.74 \leq 1.0$).

In contrast to FNO, the baseline U-Net changes by only −22% to +24% under Eq. (13). In our controlled decoder ablation (`ResizeUNet`), replacing stride-2 transposed convolutions with bilinear upsampling and 3×3 convolution collapses U-Net's TPFA face-to-cell-center

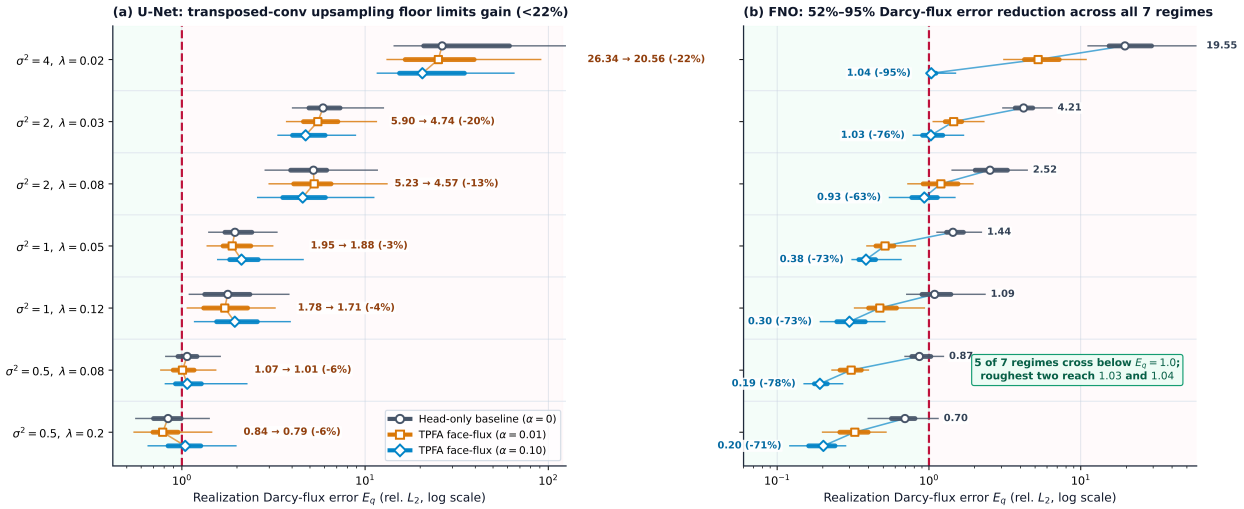


Figure 6: **Architectural response to conservative TPFA face-flux supervision across seven benchmark regimes.** Horizontal quantile forest plots (25%–75% IQR capsules, 5%–95% whiskers, and medians across $\alpha \in \{0, 0.01, 0.10\}$) for **(a)** U-Net and **(b)** FNO. FNO cuts median velocity error by 52%–73% at $\alpha = 0.01$ and 63%–95% at $\alpha = 0.10$, whereas baseline U-Net is limited by stride-2 transposed-convolution interface oscillations.

error ratio from 1.96× to 1.05× at $(1.0, 0.05)$ and from 1.65× to 1.08× at $(2.0, 0.03)$, cutting divergence imbalance $\mathcal{I}_{\text{div}}$ by 78% (5.32 → 1.19) and 73% (13.09 → 3.54). Under $\alpha = 0.01$ at $(2.0, 0.03)$, `ResizeUNet` reduces face-flux error by 22% (5.54 → 4.30) compared with −6% for the transposed-convolution baseline, confirming that transposed convolutions drive the interface amplification while FNO's global spectral modes provide stronger long-range derivative coupling (−65%).

3.5. Two-Level Transport Decoupling and Stage 2 Hybrid Multigrid Recovery

Pollock streamline tracking across 24 test realizations (984 particles per solver; Table 5 and Figure 7) reveals a two-level physical decoupling between state accuracy, global L_2 flux accuracy, and local mass conservation. At the first level, head-trained U-Net ($E_h = 5.1\%$) and head-trained FNO ($E_h = 5.2\%$) suffer near-total streamline breakdown: 0.0% (0/984) and 2.2% (22/984) of released particles reach the outflow boundary, while 73.9% and 89.4% stall inside artificial interior divergence sinks. At the second level, supervising FNO on TPFA face fluxes ($\alpha = 0.01$) cuts median face-flux error from 1.49 to 0.52 and raises particle breakthrough more than 24-fold to 54.4% (535/984 particles across 20/24 realizations) with an accurate median paired travel-time ratio of $t/t_{\text{ref}} = 1.04$. Nevertheless, residual local divergence ($\mathcal{I}_{\text{div}} = 0.23$) in the roughest realizations still traps 17.8% of trajectories and backwashes 27.8%, demonstrating that global L_2 velocity skill ($E_q < 1.0$), while necessary for accurate travel times, does not guarantee 100% streamline connectivity without pointwise divergence closure ($\nabla_h \cdot \mathbf{q}_f = 0$). Conversely, the untrained linear profile $h_0(x) = 1 - x$ achieves 100% arrival because $\partial_y h_0 \equiv 0$ eliminates interior sinks, but its 1D row locking across low-K inclusions yields a large flux error ($E_{q,\text{face}} = 1.27$) and a 1.33× slow bias in travel time.

Table 5: Pollock advective particle-tracking outcomes and initial-guess comparison across 24 held-out test realizations (984 particles per velocity source) spanning $(\sigma^2, \lambda) \in \{(0.5, 0.08), (1.0, 0.05), (2.0, 0.03)\}$, before and after Stage 2 single-cycle AMG defect correction ($\hat{h}_{\text{corr}} = \hat{h} + z_1$).

Velocity Field Source (24 cells, 984 particles)	**Med.** $E_{q,\text{face}}$	**Arrived (%)**	**Stalled (%)**	**Backwashed (%)**	**Active Cells**	**Med.** t/t_{ref} **[Note]**
Finite-volume reference (h_{ref})	0.000	984 (100.0%)	0 (0.0%)	0 (0.0%)	24/24	1.00 [exact]
Untrained linear profile ($h_0 = 1 - x$)	1.272	984 (100.0%)	0 (0.0%)	0 (0.0%)	24/24	1.33 [1.06–2.72]
U-Net (head-only, $\alpha = 0$)	3.358	0 (0.0%)	727 (73.9%)	257 (26.1%)	0/24	Undefined
FNO (head-only, $\alpha = 0$)	1.487	22 (2.2%)	880 (89.4%)	82 (8.3%)	1/24	2.42 [1 cell]
FNO (flux-weighted, $\alpha = 0.01$)	0.519	535 (54.4%)	175 (17.8%)	274 (27.8%)	20/24	1.04 [0.90–3.79]
Zero guess ($h = 0$) + 1 AMG V-cycle	0.349	123 (12.5%)	482 (49.0%)	379 (38.5%)	14/24	1.05 [incomplete]
Linear profile (h_0) + 1 AMG V-cycle	0.190	984 (100.0%)	0 (0.0%)	0 (0.0%)	24/24	1.00 [2.4× err.]
U-Net ($\alpha = 0$) + 1 AMG V-cycle	0.226	725 (73.7%)	259 (26.3%)	0 (0.0%)	24/24	1.03 [0 backwash]
FNO ($\alpha = 0$) + 1 AMG V-cycle	0.166	827 (84.0%)	140 (14.2%)	17 (1.7%)	24/24	1.00 [1.0026]
FNO ($\alpha = 0.01$) + 1 AMG V-cycle	**0.080**	**925 (94.0%)**	**51 (5.2%)**	**8 (0.8%)**	**24/24**	**1.00 [0.9982]**

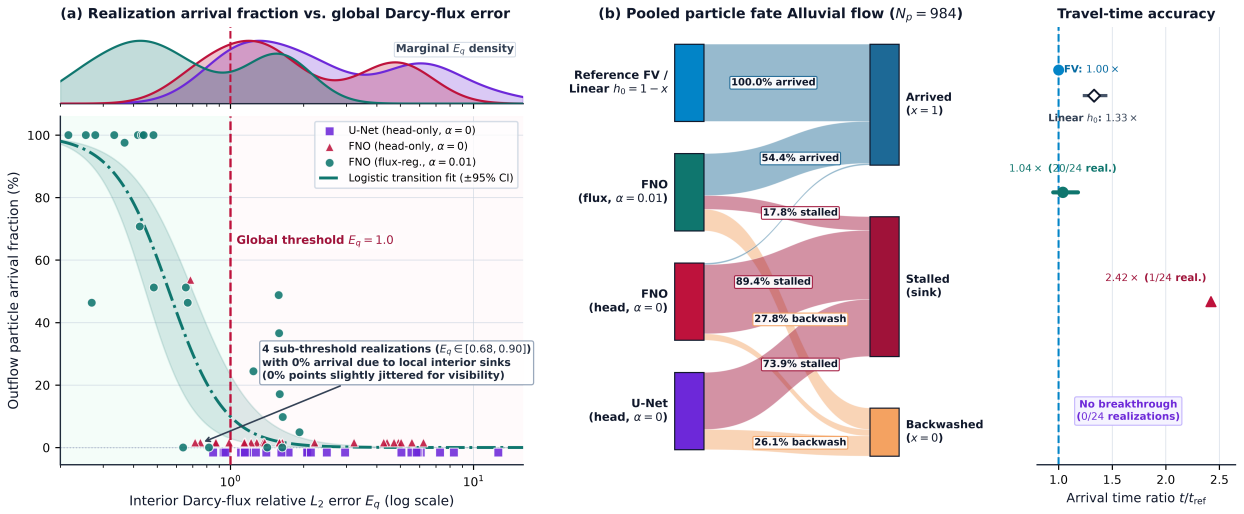


Figure 7: **Advective particle-tracking fates and travel-time accuracy** (984 **particles across** 24 **realizations). Left (a):** Outflow arrival percentage plotted against interior Darcy-flux error E_q. **Right (b):** Alluvial diagram tracing particle fates into outflow arrival, interior stalling, or inflow backwashing, paired with arrival travel-time ratios t/t_{ref}.

3.6. Stage 2: Reference-Free AMG Residual Screening, Defect Correction, and Cost–Accuracy Scaling

To screen and correct individual velocity predictions online without ground-truth solves, we exploit the exact discrete dual energy identity for the TPFA residual $r = b - A_K\hat{h}$ and true head error $e = h - \hat{h}$ (Vohralík, 2008; Ern et al., 2010; Ernst et al., 2025):

$$A_K e = r, \qquad \|\delta\mathbf{F}\|_{T^{-1}}^2 = \delta\mathbf{F}^\top T^{-1}\delta\mathbf{F} = e^\top A_K e = r^\top A_K^{-1} r, \tag{14}$$

where T is the diagonal matrix of harmonic face transmissibilities. Approximating $z_1 \approx A_K^{-1} r$ via a single smoothed-aggregation AMG V-cycle (Notay, 2010; Bell et al., 2023) yields the protected reference-free residual indicator

$$\eta_F = \frac{\|\mathbf{q}_f(z_1)\|_{2,\Omega_{\text{int}}}}{\max\left(\|\mathbf{q}_f(\hat{h})\|_{2,\Omega_{\text{int}}},\, \epsilon\right)}, \quad \epsilon = 10^{-12}, \tag{15}$$

and the one-step defect-corrected head field $\hat{h}_{\text{corr}} = \hat{h} + z_1$.

Across all 3,264 atlas evaluations, η_F ranks true interior face-flux error with Spearman $\rho = 0.92$ and separates failed ($E_{q,\text{face}} > 1.0$) from reliable predictions with ROC-AUC = 0.997, substantially outperforming the unpreconditioned residual $\|r\|_2/\|b\|_2$ ($\rho = 0.66$, dropping to $\rho = 0.47$ on FNO). The same indicator achieves ROC-AUC = 0.9998 (95% bootstrap CI $[0.9985, 1.0000]$, $n = 144$ across 3 regimes $\times 16$ realizations $\times 3$ models) on the sinuous channelized regimes, ROC-AUC = 0.9912 (95% CI $[0.9676, 1.0000]$, $n = 48$ across 16 realizations $\times 3$ models) on the MADE alluvial cross-sections, and ROC-AUC = 1.0000 (95% CI $[1.0000, 1.0000]$, $\rho = 0.955$, $n = 54$) when evaluating trained U-Net, head-FNO, and Flux-FNO models directly on refined 256×256 conductivity fields.

Applying z_1 as a single-step defect correction ($\hat{h}_{\text{corr}} = \hat{h} + z_1$) cuts pooled median face-flux error by 93% across the 128^2 Gaussian atlas ($2.83 \to 0.20$), by 90.3% on channelized media ($1.18 \to 0.115$), by 88.5% on real 256^2 neural predictions ($2.04 \to 0.235$), and restores Pollock particle arrival on the MADE alluvial cross-sections from 26.2% to 89.3% ($E_{q,\text{face}}$: $0.74 \to 0.139$). Crucially, Table 5 demonstrates that the neural operator provides genuine warm-start value: after one AMG V-cycle, 'Flux-FNO + 1 V-cycle' achieves median $E_h = 0.56\%$ and $E_{q,\text{face}} = 0.080$ (94.0% Pollock arrival across 24/24 realizations, 0.8% backwashing, travel-time ratio 1.00), which is 2.4× more accurate in face flux than 'Linear + 1 V-cycle' ($E_{q,\text{face}} = 0.190$) and 4.4× more accurate than 'Zero + 1 V-cycle' ($E_{q,\text{face}} = 0.349$, 12.5% arrival). Standalone AMG starting from $h = 0$ or $h_0(x) = 1 - x$ requires 2.0 V-cycles (double the multigrid solve work) to match the flux accuracy of 'Flux-FNO + 1 V-cycle'.

Evaluating end-to-end computational costs clarifies the operational break-even regimes between standalone Flux-FNO, hybrid 'Flux-FNO + 1-cycle AMG', and direct numerical solvers. Training Flux-FNO requires $T_{\text{train}} \approx 110$ s on 128 realizations. In moderate heterogeneity ($\sigma^2 \leq 1.0$) or channelized media where standalone Flux-FNO achieves $E_q = 0.31$–0.51 without multigrid post-processing, a single neural forward pass takes $T_{\text{neural}} = 2.8$ ms (128^2) or 33 ms (256^2) versus $T_{\text{direct}} = 41$ ms (128^2) or 133 ms (256^2), amortizing training after $\sim$ 2,800 forward evaluations (a single ensemble data-assimilation workflow). When strict local conservation is required for streamline tracking on refined grids, $T_{\text{neural}} + T_{\text{AMG setup}} + T_{\text{V-cycle}}$ requires 0.080 s versus 0.133 s for direct factorization on real 256^2 predictions (1.66× speedup; 3.6× speedup when the AMG hierarchy is reused across boundary conditions, and 3.2× total speedup on 512^2 grids at 0.42 s versus 1.34 s).

4. Discussion

Our findings resolve why subsurface neural surrogates with test head errors below 5%–6% and head $R^2 > 0.99$ frequently fail in downstream mass-balance or solute-transport modeling (Fuks and Tchelepi, 2020; Wang et al., 2020; Mamud et al., 2024). Standard L_2 head training weights all Fourier modes equally ($|\widehat{\delta h}(\mathbf{k})|^2$), whereas the Darcy velocity norm weights modes by $k^2|\widehat{\delta h}(\mathbf{k})|^2$ convolved with $\widehat{K}(\mathbf{k})$. In rough media, L_2-trained surrogates fit the dominant macro-scale pressure ramp ($k \leq 4$) while leaving a flat high-wavenumber residual floor ($k \geq 12$) that accounts for $< 1\%$ of head variance but reverses local cell-face gradients and traps 73.9%–89.4% of advective particles. Subsurface surrogate benchmarks should therefore report conservative TPFA face-flux error ($E_{q,\text{face}}$), local divergence imbalance ($\mathcal{I}_{\text{div}}$), and Pollock breakthrough statistics benchmarked against the zero-training linear profile $h_0(x) = 1 - x$.

Comparing solver architectures and training objectives highlights the interplay between operator parametrization and numerical conditioning. Whereas unsupervised finite-volume divergence losses (Yan et al., 2025) penalize a second-order difference operator ($\nabla_h \cdot (T\nabla_h \hat{h})$) that is ill-conditioned across high-wavenumber modes and discontinuous K interfaces ($\sigma^2 \geq 1.0$), direct supervision on reference TPFA face fluxes constrains a first-order difference in $\hat{h}$ (H^1_K energy norm) using global FNO modes, reducing velocity error by 52%–95% across Gaussian regimes and by 57%–73% zero-shot on channelized and MADE-type alluvial media. In convolutional U-Nets, replacing stride-2 transposed convolutions with bilinear resize-convolution eliminates the 2.26× TPFA interface error amplification, although global spectral layers remain superior to local 3×3 stencils for capturing long-range flux coupling.

Synthesizing these components yields the two-stage operational workflow summarized in Table 6. In Stage 1, practitioners evaluate σ^2/λ, σ^2, or parameter-free discrete gradient energy $\mathcal{R}_{\nabla Y}$ directly on the target conductivity ensemble to route rough regimes to conservative face-flux training. In Stage 2, online inference uses standalone Flux-FNO for fast ensemble flux estimation or pairs Flux-FNO with a single $O(N^2)$ AMG V-cycle (η_F screening and $\hat{h} + z_1$ defect correction) to guarantee local mass conservation and domain-spanning streamline transport. While validated here on two-dimensional steady-state single-phase flow across Gaussian, channelized, and alluvial lens media, the discrete TPFA face-flux loss and dual-norm AMG residual identity ($A_K e = r$) extend directly to three-dimensional hexahedral grids and implicit time-stepping schemes in transient or multiphase porous media flow.

5. Conclusions

We have developed and validated a reliability-aware two-stage computational workflow (`SubsurfaceReliability v1.0`) across 24 lognormal heterogeneity regimes (3,264 evaluations), 12 unseen off-grid regimes, and non-Gaussian channelized and MADE-type alluvial lens media. Head-trained U-Net and FNO surrogates achieve median head errors of 5.9% and 5.5% yet exceed the zero-velocity error threshold ($E_q > 1.0$) in 91.4% and 75.3% of evaluations because k^2 derivative weighting amplifies high-wavenumber head residuals across conductivity interfaces, while stride-2 transposed convolutions induce a 2.26× TPFA interface error amplification in U-Net that is eliminated by bilinear resize-convolution (1.03× to 1.08×). Prior to training, medium roughness invariants (σ^2/λ, σ^2, and parameter-free discrete gradient energy $\mathcal{R}_{\nabla Y}$) screen regime suitability with instance-level ROC-AUCs of 0.915–0.955, achieving 1.000 balanced accuracy (12/12) for U-Net and 100% sensitivity for FNO on 12 unseen off-grid regimes. Direct supervision on conservative TPFA face fluxes ($\alpha \in \{0.01, 0.10\}$) overcomes the ill-conditioning of second-order unsupervised divergence penalties, reducing FNO velocity error by 52%–95% across Gaussian regimes and by 57%–73% zero-shot on channelized and MADE-type alluvial lens media. Finally, a single AMG V-cycle provides a reference-free residual indicator η_F (ROC-AUC = 0.991–1.000 across Gaussian, channelized, alluvial, and 256^2 grids), cuts median face-flux error by 89%–93% as a one-step defect correction (2.4× to 4.4× more accurate than one AMG cycle from linear or zero guesses), and restores 94.0% Pollock streamline breakthrough with a median travel-time ratio of 1.00.

Table 6: Executable two-stage computational workflow (`SubsurfaceReliability v1.0`) for pre-training roughness screening, flux-informed operator training, and online AMG residual screening and defect correction.

Workflow Stage	Algorithmic Procedure and Operational Decision Rules
Inputs	Discretized log-conductivity ensemble $\{Y^{(m)}(\mathbf{x})\}_{m=1}^{M}$ on $N \times N$ grid ($K = \exp Y$); target solver family $\mathcal{M} \in \{\text{U-Net}, \text{FNO}, \text{PINN}\}$; conservative TPFA stencil $A_K\mathbf{h} = \mathbf{b}$.
Stage 1: Pre-Training Roughness Screening & Loss Selection	**1a. Compute medium roughness invariant ($O(N^2)$ per field):** Evaluate (σ^2, λ) or parameter-free discrete gradient energy $\mathcal{R}_{\nabla Y} = L^2\langle\lvert\nabla Y\rvert^2\rangle$. **1b. Apply calibrated regime safety screen:** Compare against baseline thresholds: $\sigma^2/\lambda < 3.33$ (U-Net), $\sigma^2 < 0.75$ (FNO), or $\sigma^2/\lambda^2 < 200$ / $\mathcal{R}_{\nabla Y} < 400$ (PINN; recalibratable via Youden's J). **1c. Select training objective:** If flagged as rough ($\geq$ threshold), train FNO with conservative TPFA face-flux supervision $\mathcal{L}(\theta) = \mathcal{L}_h + (\alpha/f_0)\mathcal{L}_q$ ($\alpha \in [0.01, 0.10]$) or solve PINN with conductivity-normalized/weighted residuals.
Stage 2: Online AMG Screening & Defect Correction	**2a. Fast neural forward pass:** Predict hydraulic head $\hat{h}$ (2.8 ms at 128^2, 33 ms at 256^2); use standalone Flux-FNO directly for ensemble flux estimation when $\sigma^2 \leq 1.0$. **2b. Single-cycle AMG dual-norm indicator ($O(N^2)$):** For rough media or transport workflows, assemble $r = b - A_K\hat{h}$, compute one AMG V-cycle $z_1 \approx A_K^{-1}r$, and evaluate $\eta_F = \lVert\mathbf{q}_f(z_1)\rVert_{2,\Omega_{\text{int}}} / \max(\lVert\mathbf{q}_f(\hat{h})\rVert_{2,\Omega_{\text{int}}}, 10^{-12})$. **2c. Reference-free screening and 1-step correction:** If η_F exceeds tolerance, apply $\hat{h}_{\text{corr}} = \hat{h} + z_1$ and reconstruct conservative face fluxes $\mathbf{q}_f(\hat{h}_{\text{corr}})$ (cutting flux error by 89%–93% and restoring 89%–94% streamline arrival).

6. Computer Code and Data Availability

- **Name of code / software:** `SubsurfaceReliability` (Version 1.0)
- **Developer:** Zijie Deng (Jiangxi Provincial Key Laboratory of Water Resources Allocation and Efficient Utilization, Jiangxi University of Water Resources and Electric Power; email: `3335196397vip@gmail.com`)
- **Year first available:** 2026
- **Program language:** Python 3.12 (with Bash automation scripts)
- **Software required:** NumPy ($\geq$ 2.0), SciPy ($\geq$ 1.14), PyTorch ($\geq$ 2.4), PyAMG ($\geq$ 5.2), pandas ($\geq$ 2.2), Matplotlib ($\geq$ 3.9), scikit-learn ($\geq$ 1.5), and h5py ($\geq$ 3.11); locked environment specifications are recorded in `environment.txt`
- **Licensing provisions:** MIT Open-Source License
- **Repository, model weights, and user documentation:** The complete source code, user guide (`USER_GUIDE.md`), quickstart tutorial (`tutorials/quickstart_workflow.py`), unit tests, archived evaluation records (`runs/` and `data/`), and all 409 trained PyTorch model checkpoints (`SubsurfaceReliability_v1.0_model_weights.tar.gz` via GitHub Releases) are openly available at https://github.com/yuelangmanle/

SubsurfaceReliability. Running `bash scripts/reproduce_figures_and_tables.sh` executes the four-gate numerical verification suite, runs the unit tests, prints all summary tables, and reproduces Figures 1–7 in approximately 15 s on a standard desktop CPU.

CRediT Authorship Contribution Statement

Zijie Deng: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Zhenyu Liao:** Data curation, Validation, Writing – review & editing. **Jiyang Yu:** Investigation, Visualization, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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